Solutions

Question 10.1:

(a) $\bar{a} = 4\hat{i} + 3\hat{j} \implies ||\bar{a}|| = \sqrt{4^2 + 3^2} = \sqrt{25} = 5$

(b) $\bar{b} = 4\hat{i} - 3\hat{j} \implies ||\bar{b}|| = \sqrt{4^2 + (-3)^2} = \sqrt{25} = 5$

(c) $\bar{c} = 3\hat{i} + 4\hat{j} + 5\hat{k} \implies ||\bar{c}|| = \sqrt{3^2 + 4^2 + 5^2} = \sqrt{25} = 7.07$

Question 10.2: For a vector to be a unit vector, we need $||\bar{v}|| = 1$ so we need to normalise the $\bar{v} = \sqrt{3^2 + 4^2 + 5^2}$ which is $||\bar{v}|| = \sqrt{50}$. Therefore the unit vector is

$$\frac{3}{\sqrt{50}}\hat{i} + \frac{4}{\sqrt{50}}\hat{j} + \frac{5}{\sqrt{50}}\hat{k}$$

Question 10.3: Provide the vector of each of the following:

(a) To construct this vector we subtract coordinates of the starting point from the ending point.

$$\vec{AB} = \langle 1-2, -3-(-7), -5-0 \rangle = \langle -1, 4, -5 \rangle$$

(b) To construct this vector we subtract coordinates of the starting point from the ending point.

$$\vec{AB} = \langle 2-1, -7-(-3), 0--(-5) \rangle = \langle 1, -4, 5 \rangle$$

Question 10.4:

1. $||\vec{a}|| = \sqrt{9 + 24 + 100} = \sqrt{134}$
2. $||\vec{p}|| = \sqrt{\frac{1}{5} + \frac{4}{5}} = 1$

Question 10.5:

- Car A will have moved $30 \times 3 = 90$ miles along the x-axis.
- Car B will have moved $60 \times 3 = 180$ miles along the direction of $3\hat{i} + 4\hat{j}$) which has magnitude 5.
- So it will have moved to $180 \times \frac{3}{5}\hat{i} + 180 \times \frac{4}{5}\hat{j}$.
- To find the distance between the two subtract one vector from the other, i.e. $90\hat{i} - 108\hat{i} - 144\hat{j}$.
- The magnitude of this vector is $\sqrt{18^2 + 144^2} = 145.1$ miles.

Question 10.6:

(a) Take partial derivatives of $f(x,y) = x^2 \sin(5y)$ (remember to treat the other variables as constant):

$$\frac{\partial f}{\partial x} = 18x \sin(5y) \qquad \frac{\partial f}{\partial y} = 45x^2 \cos(5y)$$

Then the gradient vector (∇f) can be constructed as follows:

$$\begin{aligned}\nabla f &= 18x\sin(5y)\,\hat{i} + 45x^2\cos(5y)\,\hat{j} \\ &= \left\langle 18x\sin(5y), 45x^2\cos(5y)\right\rangle\end{aligned}$$

(b) Take partial derivatives of $f(x,y,z) = z\exp(-xy)$ (remember to treat the other variables as constant):

$$\frac{\partial f}{\partial x} = -yz\exp(-xy) \quad \frac{\partial f}{\partial y} = -xz\exp(-xy) \quad \frac{\partial f}{\partial z} = \exp(-xy)$$

Then the gradient vector (∇f) can be constructed as follows:

$$\begin{aligned}\nabla f &= -yz\exp(-xy)\,\hat{i} - xz\exp(-xy)\,\hat{j} + \exp(-xy)\,\hat{k} \\ &= \langle -yz\exp(-xy), -xz\exp(-xy), \exp(-xy)\rangle\end{aligned}$$

(c) Take partial derivatives of $f(x,y) = 1\times10^{-3}(x-400)^2 + 1\times10^{-3}y^2$ (remember to treat the other variables as constant):

$$\frac{\partial f}{\partial x} = 2\times10^{-3}(x-400) \quad \frac{\partial f}{\partial y} = 2\times10^{-3}y$$

Then the gradient vector (∇f) can be constructed as follows:

$$\begin{aligned}\nabla f &= 2\times10^{-3}(x-400)\,\hat{i} + 2\times10^{-3}y\,\hat{j} \\ &= \left\langle 2\times10^{-3}(x-400), 2\times10^{-3}y\right\rangle\end{aligned}$$

Question 10.7: Take partial derivatives of $f(x,y) = 1\times10^{-3}(x-400)^2 + 1\times10^{-3}y^2$ (remember to treat the other variables as constant):

$$\frac{\partial f}{\partial x} = 2\times10^{-3}(x-400) \quad \frac{\partial f}{\partial y} = 2\times10^{-3}y$$

Then the gradient vector (∇f) can be constructed as follows:

$$\begin{aligned}\nabla f &= 2\times10^{-3}(x-400)\,\hat{i} + 2\times10^{-3}y\,\hat{j} \\ &= \left\langle 2\times10^{-3}(x-400), 2\times10^{-3}y\right\rangle\end{aligned}$$

At $(x,y) = (800,400)$ the gradient vector is:

$$\begin{aligned}\nabla f(x,y) &= \left\langle 2\times10^{-3}(800-400), 2\times10^{-3}\times400\right\rangle \\ &= \langle 0.8, 0.8\rangle\end{aligned}$$

Calculate the magnitude of the vector field:

$$\begin{aligned} ||\nabla f|| &= \sqrt{0.8^2 + 0.8^2} \\ &= 1.13 \end{aligned}$$

Question 10.8:

(a) Take partial derivatives of $f(x,y) = x^4 + 6\sqrt{y} - 10$. By taking the derivative with respect to x we hold other variables constant. In this case we hold y as a constant. The partial derivative with respect to x is then:

$$\frac{\partial f}{\partial x} = 4x^3$$

The second and third term in the expression always differentiate to 0 for x. To arrive at the partial derivative with respect to y we treat x as constant and thus:

$$\frac{\partial f}{\partial y} = \frac{3}{\sqrt{y}}$$

(b) Take partial derivatives of $f(x,y) = cos\left(\frac{4}{x}\right) e^{x^2y-5y^3}$. By taking the derivative with respect to x we hold other variables constant. Well lets start with our partial derivative with respect to x. We need to use the product rule as we have two expressions multiplied together here that contain x: a cosine function and an exponential. Split the function and differentiate:

$$\begin{aligned} u(x,y) = \cos\left(\frac{4}{x}\right), \quad &\Longrightarrow \quad \frac{\partial u}{\partial x} = -\frac{4}{x}\sin\left(\frac{4}{x}\right), \\ v(x,y) = \exp(x^2y - 5y^3) \quad &\Longrightarrow \quad \frac{\partial v}{\partial x} = 2xy\exp(x^2y - 5y^3) \end{aligned}$$

Apply the product rule:

$$\begin{aligned} \frac{\partial f}{\partial x} &= u \times \frac{\partial v}{\partial x} + v \times \frac{\partial u}{\partial x} \\ &= \frac{4}{x} sin\left(\frac{4}{x}\right)\exp(x^2y - 5y^3) + 2xy\cos\left(\frac{4}{x}\right)\exp(x^2y - 5y^3) \end{aligned}$$

For the partial derivative with respect to y, we do not need the product rule since variable y only appears in the exponential. We thus arrive at the following:

$$\frac{\partial f}{\partial y} = \left(x^2 - 15y^2\right)\cos\left(\frac{4}{x}\right)\exp(x^2y - 5y^3)$$

(c) Take partial derivatives of $z(u,v) = \frac{9u}{u^2+5v}$. By taking the derivative with respect to x we hold other variables constant. Well lets start with our partial derivative with

respect to u. We need to use the quotient rule as u appears both in the numerator and denominator. Split the function and differentiate:

$$\begin{aligned} f(u,v) = 9u, \quad &\implies \quad \frac{\partial f}{\partial u} = 9, \\ g(u,v) = u^2 + 5v \quad &\implies \quad \frac{\partial g}{\partial u} = 2u \end{aligned}$$

Apply the quotient rule:

$$\begin{aligned} \frac{\partial y}{\partial x} &= \frac{g \times \frac{\partial f}{\partial u} - f \times \frac{\partial g}{\partial u}}{g^2} \\ &= \frac{9(u^2+5v) - 9u(2u)}{(u^2+5v)^2} \\ &= \frac{-9u^2 + 45v}{(u^2+5v)^2} \end{aligned}$$

For the partial derivative with respect to v, we do not need the quotient rule since variable v only appears in the exponential. We thus arrive at the following:

$$\frac{\partial f}{\partial y} = \frac{-45u}{(u^2+5v)^2}$$

Question 10.9: Take the partial derivative of

$$f(x,y) = 100 \exp\left(-1 \times 10^{-5}\left[x^2 + y^2\right]\right)$$

with respect to x:

$$\frac{\partial f(x,y)}{\partial x} = -1 \times 10^{-5} \times 100 \times 2x \exp\left(-1 \times 10^{-5}\left[x^2 + y^2\right]\right)$$

Remember that taking the derivative with respect to x we hold other variables constant.